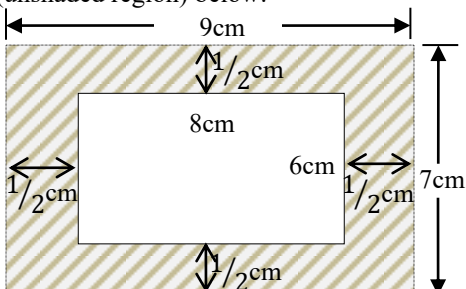


MATHS 1978 SOLUTION ANSWERS AT A GLANCE

1.A 2.No correct Option 3.B 4.B 5.E
6.C 7.A 8.B 9.C 10.E 11.D 12.E 13.-
14.A 15.D 16.E 17.D 18.B 19.C 20.D
21.C 22.E 23.E 24.- 25.B 26.E 27.B
28.A 29.C 30.C 31.B 32.B 33.C 34.A
35.C 36.D 37.B 38.B 39.A 40.B 41.D
42.E 43.A 44.E 45.D

EXPLANATION TO ANSWERS

1. A. Consider the frame (shade region) and the rectangular picture (unshaded region) below:



Note that the length of the picture is 8cm (the longest side) and the breadth is 6cm

The area of the picture is;

$$\text{Area} = \text{length} \times \text{breadth} = 8\text{cm} \times 6\text{cm} = 48\text{cm}^2$$

The area of the whole surface (i.e picture + frame) is:

$$\text{Area} = \text{length} \times \text{breadth} = (8\text{cm} + \frac{1}{2}\text{cm} + \frac{1}{2}\text{cm}) \times (6\text{cm} + \frac{1}{2}\text{cm} + \frac{1}{2}\text{cm}) = 9\text{cm} \times 7\text{cm} = 63\text{cm}^2$$

$$\text{Area of frame} = \text{area of the whole surface} - \text{area of the picture} = 63\text{cm}^2 - 48\text{cm}^2$$

$$= 15\text{cm}^2 \text{ or } 15\text{square centimeter}$$

2. No correct option.

The sum of $3\frac{7}{8}$ and $1\frac{1}{3}$ is;

$$3\frac{7}{8} + 1\frac{1}{3} = 4\frac{21+8}{24} = 4\frac{29}{24} = 4 + 1\frac{5}{24} = 5\frac{5}{24}$$

The difference between $\frac{3}{8}$ and $1\frac{2}{3}$

$$\begin{aligned} &= \frac{3}{8} - 1\frac{2}{3} \\ &= \frac{3}{8} - \frac{3}{3} \quad \text{note; } 1\frac{2}{3} = \frac{5}{3} \\ &= \frac{3-40}{24} = \frac{31}{24} = 1\frac{7}{24} \end{aligned}$$

Observe that the sum of $3\frac{7}{8}$ and $1\frac{1}{3}$ which is $5\frac{5}{24}$ is never less than the difference between $\frac{3}{8}$ and $1\frac{2}{3}$ which is $1\frac{7}{24}$, therefore there is no correct answer.

3. B. To multiply $(x + 3y + 5)(2x^2 + 5y + 2)$ we expand as follows'

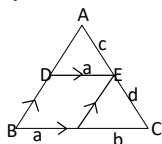
$$\begin{aligned} &= x(2x^2 + 5y + 2) + 3y(2x^2 + 5y + 2) + 5(2x^2 + 5y + 2) \\ &= 2x^3 + 5xy + 2x + 6x^2y + 15y^2 + 6y + 10x^2 + 25y + 10 \end{aligned}$$

Collecting like terms gives;

$$\begin{aligned} &= 2x^3 + 6x^2y + 5xy + 15y^2 + 6y + 25y + 10x^2 + 2x + 10 \\ &= 2x^3 + 6x^2y + 5xy + 15y^2 + 31y + 10x^2 + 2x + 10 \end{aligned}$$

Which is the required solution. Note that $x^2y = yx^2$

4. B. Generally; consider the figure below

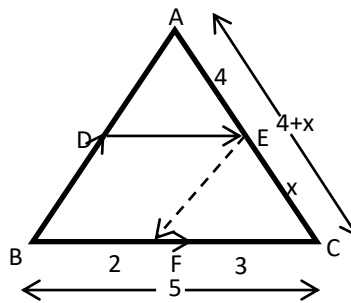


Then we have that; $\frac{|AE|}{|DE|} = \frac{|AC|}{|BC|}$ --- (i) Generally,

$$i.e \frac{c}{a} = \frac{c+d}{a+b}$$

In our case, $|DE| = |BF| = 2\text{cm}$; (parallel sides are equal)

Let $|EC| = x$, then



Applying (1) gives; $\frac{4}{2} = \frac{4+x}{5}$

On cross multiplying we have; $4 \times 5 = 2(4 + x)$

$$20 = 8 + 2x$$

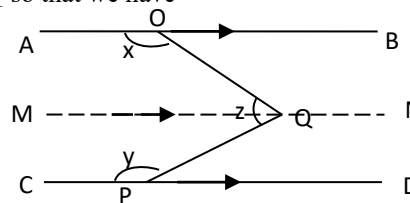
$$20 - 8 = 2x$$

$$12 = 2x$$

$$x = \frac{12}{2} = 6\text{cm}$$

Hence, the length of $|EC| = x = 6\text{cm}$

5. E. From the figure given, construct a line MN passing through Q so that we have



Observe that; $\angle AOP = x$, $\angle CPQ = y$, $\angle OQP = z$

And $\angle OQN = \angle AOP = x$ (alternating angles are equal)

$\angle PQN = \angle CPQ = y$ (alternating angles are equal)

But

$\angle OQP + \angle OQN + \angle PQN = 360^\circ$ (sum of angles at a point)

$z + x + y = 360^\circ$; Hence, $x + y + z = 360^\circ$

6. C. To obtain the ascending order of these fractions, we simply reduce the fraction into decimals using the calculator and then arranging the decimals in increasing order i.e

$$\frac{3}{5} = 0.6; \frac{9}{16} = 0.5625; \frac{34}{59} = 0.57627; \frac{71}{97} = 0.73196$$

From the decimals, the ascending order is; $\frac{9}{16}; \frac{34}{59}; \frac{3}{5}; \frac{71}{97}$

7. A. For a progression or sequence, $T_1, T_2, T_3, T_4, \dots$ to be a geometric progression, Then, $\frac{T_2}{T_1} = \frac{T_3}{T_2} = \frac{T_4}{T_3} = \dots = r$

Where r is the common ratio

The n th term of the geometric progression is; $T_n = ar^{n-1}$

Where n is the number of terms, a is the first term; r is the common ratio; T_n is the n th term

The sum of the first n th term of the geometric progression is

Either $S_n = \frac{a(1-r^n)}{1-r}$ if $|r| < 1$ Or $S_n = \frac{a(r^n-1)}{r-1}$ if $|r| > 1$

The sum to infinity of the geometric progression is $S_\infty = \frac{a}{1-r}$

In our case; $T_1 = 1, T_2 = x, T_3 = x^2, T_4 = x^3$

$$i.e \frac{T_2}{T_1} = \frac{x}{1} = \frac{T_3}{T_2} = \frac{x^3}{x^2} = x$$

this shows that the progression is a geometric progression with ratio $r = x$.

The first term of the geometric progression is $T_1 = 1 = a$

The sum to infinity of this progression is $S_\infty = \frac{a}{1-r} = \frac{1}{1-x}$

Hence, $S_{\infty} = \frac{1}{1-x}$

8. **B.** This is variation; a
When x varies directly as y ,
We write; $x \propto y$ i.e $x = Ky$; where K is the constant of proportionality
when x varies inversely as y , we write ; $x \propto \frac{1}{y}$
i.e $x = \frac{K}{y}$; where K is the constant of proportionality.

In our case ; $N \propto P_A P_B$ and $N \propto \frac{1}{D^2}$

On combining, we have ; $N \propto \frac{P_A P_B}{D^2} \rightarrow N = \frac{K P_A P_B}{D^2}$ where K is the constant of proportionality

9. **C.** Let the square root of $170 - 20\sqrt{30}$ be $\sqrt{x} - \sqrt{y}$,

Then, $\sqrt{170 - 20\sqrt{30}} = \sqrt{x} - \sqrt{y}$

On squaring both sides

We have; $170 - 20\sqrt{30} = (\sqrt{x} - \sqrt{y})^2$

$= (\sqrt{x} - \sqrt{y})(\sqrt{x} - \sqrt{y})$

$= (\sqrt{x})^2 - \sqrt{x}\sqrt{y} - \sqrt{y}\sqrt{x} + (\sqrt{y})^2 = x - 2\sqrt{xy} + y$

i.e $170 - 20\sqrt{30} = x + y - 2\sqrt{xy}$

On comparing both sides we have

$170 = x + y \quad \text{--- (1)}$

$-20\sqrt{30} = -2\sqrt{xy} \quad \text{--- (2)}$

Dividing through by -2

Gives $10\sqrt{30} = \sqrt{xy}$ i.e $\sqrt{xy} = 10\sqrt{30}$

$(\sqrt{xy})^2 = (10\sqrt{30})^2 = 10^2(\sqrt{30})^2 = 100(30) = 3000$

So that $xy = 3000 \quad \text{--- (3)}$

$x + y = 170 \quad \text{--- (1)}$

From (1); $y = 170 - x \quad \text{--- (4)}$, substitute (4) in (3)

Gives $x(170 - x) = 3000$

$170 - x^2 = 3000$

$x^2 - 170x + 3000 = 0$

Factorisation : $(x^2)(3000) = 3000x^2$

Product : $(-150x)(-20x) = 3000x^2$

Sum : $(-150x) + (-20x) = -170x$

So that we have; $x^2 - 150x - 20x + 3000 = 0$

$x(x - 150) - 20(x - 150) = 0$

$(x - 150)(x - 20) = 0$

$x - 150 = 0$ or $x - 20 = 0$

$\rightarrow x = 150$ or $x = 20$

From (4), $y = 170 - x$

When $x = 150$; $y = 170 - 150 = 20$

When $x = 20$; $y = 170 - 20 = 150$

Since we have that ; $\sqrt{170 - 20\sqrt{30}} = \pm(\sqrt{x} - \sqrt{y})$

Then $x > y$; $\rightarrow$ we choose $x = 150, y = 20$

And $\sqrt{170 - 20\sqrt{30}} = \pm(\sqrt{150} - \sqrt{20})$

$= \pm(\sqrt{25 \times 6} - \sqrt{4 \times 5})$

$= \pm(5\sqrt{6} - 2\sqrt{5})$

Hence the answer is $+(5\sqrt{6} - 2\sqrt{5})$ or $-(5\sqrt{6} - 2\sqrt{5})$

$5\sqrt{6} - 2\sqrt{5}$ or $-5\sqrt{6} + 2\sqrt{5}$

$5\sqrt{6} - 2\sqrt{5}$ or $2\sqrt{5} - 5\sqrt{6}$

10. **E.** $x^2 + 4 = 0$; $x^2 = -4$

$x = \pm\sqrt{-4}$

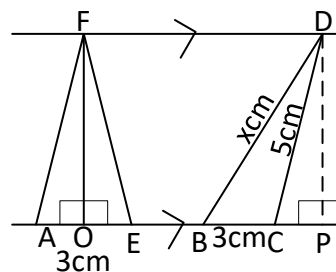
But $i = \sqrt{-1}$ (i.e i is complex number)

So that. $x = \pm\sqrt{4x(-1)} = \pm(\sqrt{4} \times \sqrt{-1})$

$= \pm(2 \times i) = \pm 2i$

Hence $x = \pm 2i$

11. **D.** Consider the modified figure below



Where $|FO| = |DP|$, the height of the figure

Considering triangle ΔAEF ,

Area of triangle is $\frac{1}{2}(\text{base} \times \text{height})$

Area of triangle = $\frac{1}{2}(|AE| \times |FO|)$

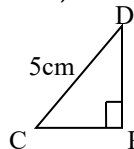
But area of AEF has been given as $6\text{sq.cm} = 6\text{cm}^2$

And $|AE| = 3\text{cm}$ Then, $6\text{cm}^2 = \frac{1}{2} \times 3\text{cm} \times |FO|$

$2 \times 6\text{cm}^2 = 3\text{cm} \times |FO|$

$|FO| = \frac{12\text{cm}^2}{3\text{cm}} = 4\text{cm}$

Let $|BD| = x\text{cm}$; Considering ΔDCP ;



This is the right angled triangle and using pythagoras' theorem, we have that

$(\text{hypotenuse side})^2 = (\text{opposite side})^2 + (\text{adjacent side})^2$

$(5\text{cm})^2 = |DP|^2 + |CP|^2$

But $|DP| = |FO| = 4\text{cm}$., (height of the figure),

Then $25\text{cm}^2 = (4\text{cm})^2 + |CP|^2$

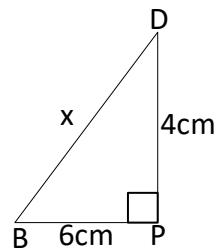
$|CP|^2 = 25\text{cm}^2 - 16\text{cm}^2 = 9\text{cm}^2$

$|CP| = \sqrt{9\text{cm}^2} = 3\text{cm}$

But $|BP| = |BC| + |CP|$

$|BP| = 3\text{cm} + 3\text{cm} = 6\text{cm}$

Now considering ΔBDP ,



Using pythagoras theorem

We have that $|BD|^2 = |BP|^2 + |DP|^2$

$|BD|^2 = (6\text{cm})^2 + (4\text{cm})^2$

$= 36\text{cm}^2 + 16\text{cm}^2 = 52\text{cm}^2$

$|BD| = \sqrt{52\text{cm}^2} = \sqrt{52}\text{cm} = \sqrt{4 \times 13}\text{cm}$

$= 2\sqrt{13}\text{cm}$

Hence, the length of BD is $= 2\sqrt{13}\text{cm}$

12. **E.** Let the number be x , then

$\log_{10} x = \bar{2}.3482$

This implies that; $x = 10^{\bar{2}.3482} = 10^{-2} \times 10^{0.3482}$

Which means that we find the antilogarithm of 0.3482, and we multiply by 10^{-2} which gives $x = 10^{-2} \times 2229 = 0.02229$

hence, the number is 0.02229

13. Let the current (now) age of the father be x
And let the current (now) age of the son be y .
Then, five (5) years ago, their ages were $(x - 5)$ and $(y - 5)$ respectively.

But, 5 years ago $(x - 5) = 3(y - 5) \dots (1)$

Now $x + y = 110 \dots (2)$

From (1); $x - 5 = 3y - 15$

$x - 3y = -15 + 5 = -10$

$x - 3y = -10 \dots (3)$

From (2) $x + y = 110$

$y = 110 - x \dots (4)$ Substituting (4) into (3)

Gives $x - 3(110 - x) = -10$

$x - 330 + 3x = -10$

$x + 3x = -10 + 330$

$4x = 320$

$x = \frac{320}{4} = 80$

Hence, the present age of the father is 80 years.

14. A. $y = 2x^2 + 9x - 35$

Since $y < 0$,

Then $2x^2 + 9x - 35 < 0$

This is a quadratic inequality and we solve as follows;

$(2x^2)(-35) = -70x^2$

product: $(-5x)(14x) = -70x^2$

sum: $(-5x) + (14x) = 9x$

Therefore, $2x^2 - 5x + 14x - 35 < 0$

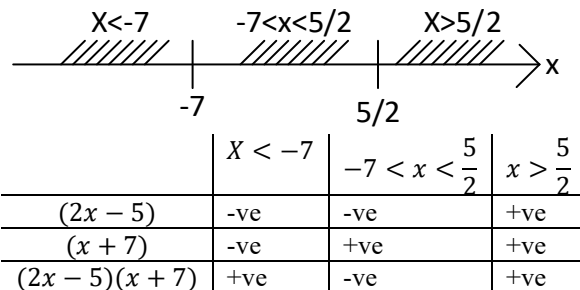
$x(2x - 5) + 7(2x - 5) < 0$

$(2x - 5)(x + 7) < 0$

For equality, we have that $2x - 5 = 0$ or $x + 7 = 0$

$x = \frac{5}{2}$ or $x = -7$

Since it is inequality, we continue further as follows :



Since the inequality is < 0 , it implies that the product of $(2x - 5)$ and $(x + 7)$ must be negative (-ve), so we choose the (-ve) products on the table, which is

$-7 < x < \frac{5}{2}$

Hence, $-7 < x < \frac{5}{2}$

15. D. Let x represent the original quantity of the food bought

Let y represent the original cost of living

The original family food bill is $x \times y = xy$

Now, the current or present quantity is reduced by 10% which implies that the present quantity is;

Present = $(100 - 10)\%$ of original quantity

$= 90\%$ of x ; $= \frac{90}{100}x$

The present cost of living is increased by 15% which implies that the present cost of living is

$(100 + 15)\%$ of original cost = 115% of y ; $= \frac{115}{100}y$

The current family food bill = quantity $\times$ current cost

$\frac{90x}{100} \times \frac{115y}{100} = \left(\frac{90}{100} \times \frac{115}{100}\right)xy = \frac{207}{200}xy$

The fractional increase in the family's food bill is now; = present food bill - original food bill

$= \left(\frac{207}{200} - \frac{1}{1}\right)xy = \frac{7}{200}xy$

Hence, the fractional increase in the food bill is $\frac{7}{200}$

16. E. $a * b = ab + a + b \dots (1)$

$a^{\circ}b = a + b + 1 \dots (2)$

Now, to find $(a * b)^{\circ}(a * c)$; we use (1) & (2) and proceed as follows;

$(a * b)^{\circ}(a * c) = (a * b) + (a * c) + 1$ using (2)

$= ab + a + b + ac + a + c + 1$

$= ab + ac + 2a + b + c + 1$

17. D. Circumference of a circle $C = 2\pi r$

where r is the radius of the circle

Surface area of a circle $A = \pi r^2$

Now the ratio of the circumference is 2:5

Let r_1 be the radius of the original disc

Let r_2 be the radius of the trimmed disc

Then $\frac{C_1}{C_2} = \frac{2\pi r_1}{2\pi r_2} = \frac{2}{5} \Rightarrow \frac{r_1}{r_2} = \frac{2}{5}$

Now, the ratio of the surface area is;

$\frac{A_1}{A_2} = \frac{\pi r_1^2}{\pi r_2^2} = \frac{r_1^2}{r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{2}{5}\right)^2 = \frac{4}{25} = 4:25$

18. B. The sum of interior angles in a polygon of n sides is

$S_n = (n - 2) \times 180^{\circ}$

For a quadrilateral, having four sides, $x = 4$,

Sum of interior angles is; $S_4 = (4 - 2) \times 180^{\circ}$

$= 2 \times 180^{\circ} = 360^{\circ}$

Sum of interior angles given is

$S_4 = (P + 10)^{\circ} + (p - 30)^{\circ} + (2p - 45)^{\circ} + (p + 15)^{\circ} = 360^{\circ}$

$P + P + 2P + P - 50^{\circ} = 360^{\circ}$

$5P = 360^{\circ} + 50^{\circ}$

$5P = 410^{\circ} \Rightarrow P = 82^{\circ}$

19. C. $\frac{a-b}{a+b} - \frac{a+b}{a-b} = \frac{(a-b)^2 - (a+b)^2}{(a+b)(a-b)}$

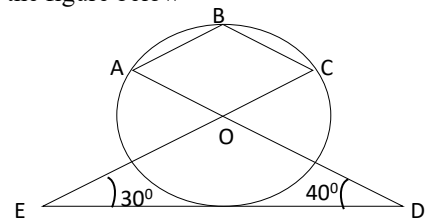
$= \frac{a^2 - 2ab + b^2 - [a^2 + 2ab + b^2]}{(a+b)(a-b)}$

$= \frac{a^2 - b^2}{a^2 - b^2}$

$= \frac{a^2 - 2ab + b^2 - a^2 - 2ab - b^2}{a^2 - b^2}$

$= \frac{-4ab}{a^2 - b^2}$

20. D. Consider the figure below



considering triangle EOD ;

$\angle OED + \angle EOD + \angle EDO = 180^{\circ}$ (sum of angles in a triangle)

$30^{\circ} + 40^{\circ} + \angle EOD = 180^{\circ}$

$\angle EOD = 180^{\circ} - 70^{\circ}$

$\angle EOD = 110^{\circ}$

considering straightline EOC ;

$\angle EOD + \angle COD = 180^{\circ}$ [sum of angles on a straight line]

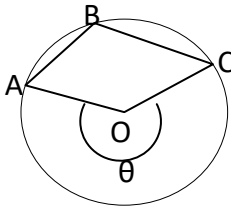
$110^{\circ} + \angle COD = 180^{\circ}$

$\angle COD = 180^{\circ} - 110^{\circ}$

$\angle COD = 70^{\circ}$

$\angle AOE = \angle COD = 70^{\circ}$ {alternating angles are equal}

considering the major sector AOC i.e



$$\text{reflex } \angle AOC = \angle AOE + \angle EOD + \angle COD$$

$$= 70^\circ + 110^\circ + 70^\circ$$

$$= 140^\circ + 110^\circ = 250^\circ$$

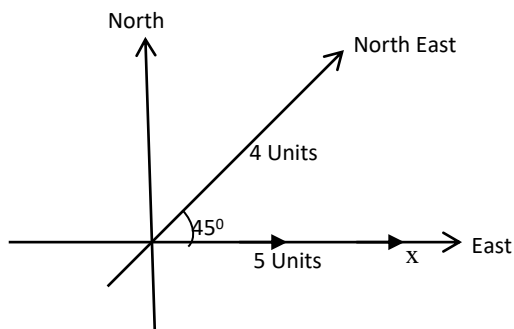
But Reflex angle $\angle AOC = 2\angle ABC$ {angle at the center is twice the angle at the circumference}

$$250^\circ = 2\angle ABC$$

$$\angle ABC = \frac{250^\circ}{2} = 125^\circ$$

$$\text{Hence } \angle ABC = 125^\circ$$

21. C. The diagram of the forces is as follows



We resolve the 4 unit force to the vertical and horizontal

$$\text{i.e } F_{\text{vertical}} = 4 \text{ units} \sin \theta = 4 \text{ units} \sin 45^\circ$$

$$= 4 \cdot \frac{\sqrt{2}}{2} \text{ units}$$

$$= 2\sqrt{2} \text{ units}$$

$$F_{\text{horizontal}} = 5 \text{ units} + 4 \text{ units} \cos \theta = 5 + 4 \cos 45^\circ$$

$$= \left(5 + \frac{4\sqrt{2}}{2}\right) \text{ units}$$

$$= (5 + 2\sqrt{2}) \text{ units}$$

$$\text{Resultant force} = \sqrt{F_{\text{vertical}}^2 + F_{\text{horizontal}}^2}$$

$$= \sqrt{(2\sqrt{2})^2 + (5 + 2\sqrt{2})^2}$$

$$= \sqrt{4(2) + (5 + 2\sqrt{2})(5 + 2\sqrt{2})}$$

$$= \sqrt{8 + (25 + 20\sqrt{2} + 4(2))}$$

$$= \sqrt{8 + (25 + 8 + 20\sqrt{2})}$$

$$= \sqrt{8 + (33 + 20\sqrt{2})}$$

$$\text{resultant force} = (\sqrt{41 + 20\sqrt{2}}) \text{ units force}$$

22. E. The minimum point of the curve is the point where

$$\frac{d^2y}{dx^2} > 0$$

The maximum point of the curve is the point where

$$\frac{d^2y}{dx^2} < 0$$

To get the stationary points to use in detecting this points

$$\text{we set } \frac{dy}{dx} = 0$$

$$\text{In our case; } y = x^2 - 5x + 5$$

$$\frac{dy}{dx} = 2x - 6; \Rightarrow \text{at } \frac{dy}{dx} = 0;$$

$$\text{we have } 2x - 6 = 0; \Rightarrow 2x = 6, x = 3$$

$$\text{Now, } \frac{dy}{dx} = 2 > 0,$$

which implies that the curve $y = x^2 - 6x + 5$

is a minimum curve with minimum point $x = 3$ and minimum value

$$y = f(x - 3) = 3^2 - 6(3) + 5$$

$$= 9 - 18 + 5 = 9 - 13 = -4$$

so the minimum point is at $(x, y) = (3, -4)$

$$23. \text{ E. } 3x - \left(\frac{1}{4}\right)^{-\frac{1}{2}} > \frac{1}{4} - x$$

$$\text{From indices, } \left(\frac{a}{b}\right)^{-\frac{1}{n}} = \left(\frac{b}{a}\right)^{\frac{1}{n}} = \sqrt[n]{\frac{b}{a}}$$

$$\text{So that we have } 3x - \left(\frac{4}{1}\right)^{\frac{1}{2}} > \frac{1}{4} - x$$

$$3x + x - \sqrt[2]{4} > \frac{1}{4}$$

$$4x - 2 - \frac{1}{4} > 0$$

$$4x - \frac{9}{4} > 0$$

$$4x > \frac{9}{4}$$

$$x > \frac{9}{4} + 4$$

$$x > \frac{9}{16}$$

24. Insufficient question

25. B. Let x be his average speed for the first 4km

$$\text{average speed} = \frac{\text{distance}}{\text{time}} \quad \text{then } x = \frac{4\text{km}}{t_1}$$

$$\Rightarrow t_1 = \frac{4}{x} \text{ hours}$$

Where t_1 is the time to run the first 4km

$$\text{Also, } (x + 2) = \frac{5\text{km}}{t_2}$$

Where t_2 is the time to run the remaining distance (5km) at $(x + 2)\text{km/hr}$ speed,

since he now run 2km/hr faster than before

$$\Rightarrow t_2 = \frac{5}{(x+2)} \text{ hours}$$

If the whole distance took him one (1) hour, then ;

$$t_1 + t_2 = 1$$

$$\frac{4}{x} + \frac{5}{x+2} = 1$$

$$\frac{4(x+2) + 5x}{x(x+2)} = 1$$

$$4(x+2) + 5x = x(x+2)$$

$$4x + 8 + 5x = x^2 + 2x$$

$$x^2 + 2x = 9x + 8$$

$$x^2 + 2x - 9x - 8 = 0$$

$$x^2 - 7x - 8 = 0$$

This is a quadratic equation and we solve as follows ;

$$x^2 - 8x + 1x - 8 = -8x^2;$$

$$\text{factorization } (-8x)(+1x) = -8x^2$$

$$(-8x) + (+1x) = -7x$$

$$x^2 - 8x + 1x - 8 = 0;$$

$$x(x - 8) + 1(x - 8) = 0$$

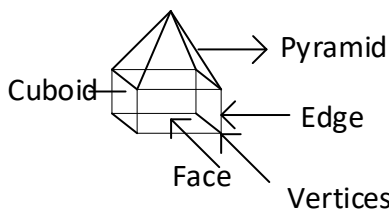
$$(x - 8)(x + 1) = 0;$$

$$x - 8 = 0 \quad \text{or} \quad x + 1 = 0$$

$$x = 8 \text{ or } x = -1$$

Hence, the average speed for the first 4km distance is 8km/hr. we did not choose -1 because, speed should not be negative

26. E.



The figure has ; 9 face ; 9 vertices : 16 edges

27. B. Kindly check solution to Question 7 before proceeding.

The n th term of a geometric progression is $T_n = ar^{n-1}$ where a is the first term, r is the common ratio and n is the number of terms.

The sum of first n th terms in a geometric progression a ;

$$S_n = \frac{a(r^n - 1)}{r - 1} \text{ if } |r| > 1 \text{ and } S_n = \frac{a(1 - r^n)}{1 - r} \text{ if } |r| < 1$$

$$\text{Now, } T_1 = a = 153, \quad T_6 = \frac{17}{27}$$

$$\text{from, } T_6 = ar^{6-1} = ar^5$$

$$T_6 = ar^5 = \frac{17}{27} \text{ --- (1)}$$

$$T_1 = a = 153 \text{ --- (2) substituting (2) into (1) gives}$$

$$153 \cdot r^5 = \frac{17}{27}; \quad r^5 = \frac{17}{27} \div 153$$

$$r^5 = \frac{17}{27 \times 153}; \quad r^5 = \frac{1}{27 \times 9}$$

$$= \frac{1}{3^3 \times 3^2} = \frac{1}{3^5} = \left(\frac{1}{3}\right)^5$$

$$r^5 = \left(\frac{1}{3}\right)^5; \quad r = \frac{1}{3}$$

the common ratio is $\frac{1}{3}$ which is less than 1

$$\text{i.e. } \frac{1}{3} < 1; \therefore a = 153,$$

$r = \frac{1}{3}$ so the sum of the first four (4) terms is;

$$\text{from } S_4 = \frac{(1-r^4)}{1-r} = \frac{153(1-(\frac{1}{3})^4)}{1-\frac{1}{3}} = \frac{153(1-\frac{1}{81})}{\frac{2}{3}}$$

$$= 153 \left(1 - \frac{1}{81}\right) \div \frac{2}{3} = 153 \left(1 - \frac{1}{81}\right) \times \frac{3}{2}$$

$$= \frac{153 \times 3}{2} \cdot \left(\frac{1}{1} - \frac{1}{81}\right); = \frac{459}{2} \cdot \left(\frac{80}{81}\right)$$

$$S_4 = \frac{459}{2} \left(\frac{80}{81}\right) = \frac{153}{1} \left(\frac{40}{27}\right) = \frac{51 \cdot (40)}{9} = \frac{170 \times 4}{3} = \frac{680}{3}$$

Hence, the sum of the first four terms is $\frac{680}{3}$

28. A. The n th term for an arithmetic progression $T_1, T_2, T_3, \dots$ with common difference as

$$d = T_2 - T_1 = T_3 - T_2 = T_4 - T_3, \dots \text{ is defined as}$$

$$T_n = a + (n-1)d \text{ --- (1)}$$

where a is the first term T_1 , d is the common difference

n is the number of terms

the sum of the first n th term of an arithmetic progression is

$$S_n = \frac{n}{2}(2a + (n-1)d) \text{ --- (2) or } S_n = \frac{n}{2}(a + L)$$

where L is the last term and a, n and d as defined before now,

$$T_1 = a = 11 \text{ --- (3)}$$

$$T_4 = 32 \text{ from (1),}$$

$$T_4 = a + (4-1)d = a + 3d$$

$$T_4 = a + 3d = 32 \text{ --- (4) substituting (3) into (4) gives}$$

$$11 + 3d = 32, \quad 3d = 32 - 11$$

$$3d = 21; \quad d = \frac{21}{3} = 7$$

$\therefore a = 11, d = 7$ now, considering formula (2) the sum of the first nine (9) terms is

$$S_9 = \frac{9}{2}(2a + (9-1)d)$$

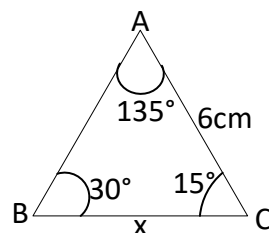
$$S_9 = \frac{9}{2}(2(11) + (9-1)7)$$

$$S_9 = \frac{9}{2}(22 + 8(7)); \quad \frac{9}{2}(22 + 56)$$

$$= \frac{9}{2} \times 78; = 9 \times 39 \quad S_9 = 351$$

A

29. C.



Let the side opposite angle 135° be x shown above, then using sine rule ; $\frac{|BC|}{\sin A} = \frac{|AC|}{\sin B} = \frac{|AB|}{\sin C}$ i.e. $\frac{x}{\sin 135^\circ} = \frac{6\text{cm}}{\sin 30^\circ}$

$$\text{note: } \sin 135^\circ = \sin(180^\circ - 45^\circ)$$

$$= \sin 45^\circ = \frac{\sqrt{2}}{2}$$

$$\sin 30^\circ = \frac{1}{2}$$

$$\text{so that we have } \frac{x}{\frac{\sqrt{2}}{2}} = \frac{6}{\frac{1}{2}}$$

$$x \div \frac{\sqrt{2}}{2} = 6 \div \frac{1}{2};$$

$$x \times \frac{2}{\sqrt{2}} = 6 \times \frac{2}{1}$$

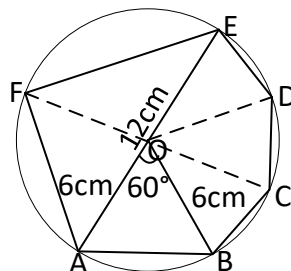
$$\frac{2x}{\sqrt{2}} = \frac{12}{1} \text{ on cross-multiplying we have,}$$

$$2x = 12\sqrt{2}$$

$$x = \frac{12\sqrt{2}}{2} = 6\sqrt{2}\text{cm}$$

Hence, the opposite side to angle 135° is $6\sqrt{2}\text{cm}$

30. C.



Diameter = 12cm

$$\text{Radius} = \frac{\text{diameter}}{2} = \frac{12\text{cm}}{2} = 6\text{cm}$$

The area of the hexagon is 6 times the area of each triangle in the circle. Since the sum of the angles at a point is 360°

$$\text{then, } 6 \cdot \angle AOB = 360^\circ \ll \angle AOB = \frac{360^\circ}{6} = 60^\circ$$

so that each triangle has an angle of 60° , this is so because the hexagon is regular

$$\text{area of triangle } \Delta AOB = \frac{1}{2}r^2 \sin \theta$$

$$\text{area of triangle } \Delta AOB = \frac{1}{2}(6\text{cm})^2 \times \sin 60^\circ = \frac{36}{2} \times \frac{\sqrt{3}}{2} = 9\sqrt{3}\text{cm}^2$$

$$\text{the area of the hexagon} = 6 \times \text{area of } \Delta AOB$$

$$= 6 \times 9\sqrt{3}\text{cm}^2 = 54\sqrt{3}\text{cm}^2$$

31. B. We first arrange the data (i.e the numbers) in ascending order as follows

$$0, 0, 0, 0, 1, 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 4, 4, 5$$

Now, the mean is $\bar{x} = \frac{\sum_{i=1}^N x_i}{N}$ where $\sum x_i$ is the sum of the data and N is the total number of data considered

$$\bar{x} = \frac{0+0+0+0+1+1+1+1+1+2+2+2+2+3+3+3+4+4+5}{20}$$

$$= \frac{6+8+9+8+5}{20} = \frac{36}{20} = 1.8$$

The median is the middle values after having arranged the data in ascending order if N is odd. But, when N is even, the median number is the sum of the 2 middle values divided by 2. In our case N = 20 (even) so that

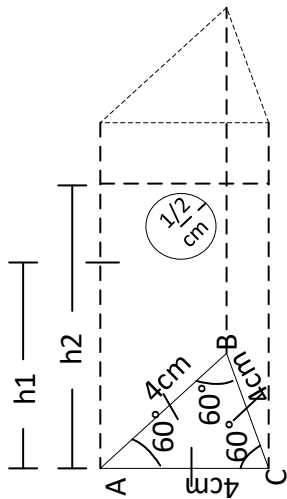
$$\text{The median is} = \frac{1+2}{2} = \frac{3}{2} = 1.5$$

The mode is the particular data that occur the most which is 1

The mean, median, mode are respectively 1.8, 1.5 and 1

32. B. $\sec^2\theta + \tan^2\theta = 3$ note that $\sin^2\theta + \cos^2\theta = 1$
 $1 + \cot^2\theta = \operatorname{cosec}^2\theta$; and $\tan^2\theta + 1 = \sec^2\theta$
 where $\sec\theta = \frac{1}{\cos\theta}$, $\operatorname{cosec}\theta = \frac{1}{\sin\theta}$ and $\cot\theta = \frac{1}{\tan\theta}$
 since $\tan^2\theta + 1 = \sec^2\theta$ then $\sec^2\theta + \tan^2\theta = 3$
 becomes $\tan^2\theta + 1 + \tan^2\theta = 3$
 $2\tan^2\theta = 3 - 1$
 $2\tan^2\theta = 2$; $\tan^2\theta = 1$
 $\tan\theta = \sqrt{1}$; $\tan\theta = \pm 1$; $\theta = \tan^{-1}(\pm 1)$
 $\theta_1 = \tan^{-1}(+1) = 45^\circ$ or $(180^\circ + 45^\circ) = 225^\circ$
 $\theta_2 = \tan^{-1}(-1) = (180^\circ - 45^\circ) = 135^\circ$
 or $(360^\circ - 45^\circ) = 315^\circ$
 θ can be $\{45^\circ, 135^\circ, 225^\circ \text{ or } 315^\circ\}$ best we choose $\theta = 45^\circ$

33. C.



Volume of a prism = base area $\times$ height

Let h_1 be the original level of water in the prism and

Let $h_2 = h_1 + x$ be the new level of water in the prism after immersing the sphere, where x is the rise of water in the prism

Base area = area of the equilateral triangle ABC, which is obtained as follows;

Area of equilateral triangle is

$$= \frac{1}{2} ab \sin\theta = \frac{1}{2} 4\text{cm} \times 4\text{cm} \times \sin 60^\circ = \frac{16}{2} \times \frac{\sqrt{3}}{2} = 4\sqrt{3}\text{cm}^2$$

Original volume of the prism is = area of equilateral triangle base $\times$ height = $4\sqrt{3}\text{cm}^2 \times h_1\text{cm} = 4\sqrt{3}h_1\text{cm}^3$

New volume of prism after immersing the sphere = original volume of prism + volume of sphere

Volume of sphere $\frac{4}{3}\pi r^3$ where r is the radius

$$= \frac{4}{3} \times \pi \times \left(\frac{1}{2}\text{cm}\right)^3 = \frac{4\pi}{3} \times \frac{1}{8}\text{cm}^3 = \frac{\pi}{6}\text{cm}^3$$

New volume of prism is now = $\left(4\sqrt{3}h_1 + \frac{\pi}{6}\right)\text{cm}^3$

Also new volume of prism is = $4\sqrt{3} \times h_2$

$$= 4\sqrt{3} \times (h_1 + x)\text{cm}^3$$

$$\Rightarrow 4\sqrt{3}h_1 + \frac{\pi}{6} = 4\sqrt{3}(h_1 + x)$$

$$\Rightarrow 4\sqrt{3}h_1 + \frac{\pi}{6} = 4\sqrt{3}h_1 + 4\sqrt{3}x$$

which gives $\frac{\pi}{6} = 4\sqrt{3}x$

$$x = \frac{\pi}{6} \div 4\sqrt{3}$$

$$x = \frac{\pi}{6} \times \frac{1}{4\sqrt{3}}$$

$$x = \frac{\pi}{24\sqrt{3}} \text{ hence, the rise of water is } \frac{\pi}{24\sqrt{3}}\text{cm}$$

$$34. \text{ A. } x^2 - y - 1 = 0 \quad \text{--- (1)}$$

$$y - 2x + 2 = 0 \quad \text{--- (2)}$$

this is a simultaneous equation and we solve by substitution method as follows ;

$$\text{from equation (2); } y = 2x - 2 \quad \text{--- (3)}$$

we then substitute equation (3) into (1)

$$x^2 - y - 1 = 0$$

$$x^2 - (2x - 2) - 1 = 0$$

$$x^2 - 2x + 2 - 1 = 0$$

$$x^2 - 2x + 1 = 0$$

this is a quadratic equation and we solve it as follows;

$$(x^2)(+1) = 1x^2$$

$$(-1x)(-1x) = (1x^2)$$

$$(-1x) + (-1x) = -2x$$

$$x^2 - 1x - 1x + 1 = 0$$

$$x(x - 1) - 1(x - 1) = 0$$

$$(x - 1)(x - 1) = 0$$

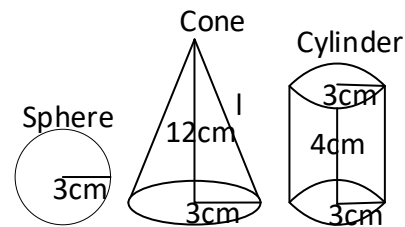
$$(x - 1) = 0 \text{ or } (x - 1) = 0 \therefore x = 1 \text{ (twice)}$$

Now we proceed to find y . but from (3), $y = 2x - 2$

when $x = 1$, $y = 2(1) - 2 = 2 - 2 = 0$

$\therefore x = 1$, $y = 0$ i.e $x, y = (1, 0)$

35. C.



$$\text{Volume of sphere} = \frac{4}{3}\pi r^3 = \frac{4}{3} \times (\pi) \times (3\text{cm})^3$$

$$= \frac{4}{3} \pi \times 27 = 4 \times 9 \times \pi = 36\pi\text{cm}^3$$

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3} \pi \times (3\text{cm})^2 \times 12\text{cm}; = \frac{1}{3} \pi \times 9 \times 12$$

$$= 4 \times 9\pi = 36\pi\text{cm}^3$$

$$\text{Volume of cylinder} = \pi r^2 h; = \pi (3\text{cm})^2 \times 4\text{cm}$$

$$= \pi \times 9\text{cm}^2 \times 4\text{cm} = 36\pi\text{cm}^3$$

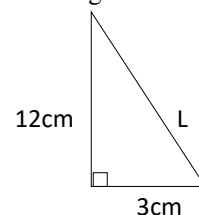
The three bodies have the same volume of $36\pi\text{cm}^3$

$$\text{The total surface area of sphere} = 4\pi r^2$$

$$= 4 \times \pi \times (3\text{cm})^2 = 4(9) \times \pi = 36\pi\text{cm}^2$$

$$\text{The total surface area of cone} = \pi r l + \pi r^2$$

Where l is the slant height and it is obtained as follows



using pythagoras theorem; $l^2 = (12\text{cm})^2 + (3\text{cm})^2$

$$l^2 = 144\text{cm}^2 + 9\text{cm}^2 = 153\text{cm}^2$$

$$l = \sqrt{153\text{cm}^2} = 3\sqrt{17}\text{cm}$$

The total surface area of cone = $\pi rl + \pi r^2$

$$= \pi \times 3\text{cm} \times 3\sqrt{17}\text{cm} + \pi \times (3\text{cm})^2$$

$$= 9\sqrt{17}\pi + 9\pi$$

$$= 9\pi(\sqrt{17} + 1)\text{cm}^2$$

The total surface area of cylinder = $2\pi rh + 2\pi r^2$

$$= 2\pi r(h + r)$$

$$= 2\pi(3\text{cm})(4\text{cm} + 3\text{cm})$$

$$= 6\pi(7) = 42\pi\text{cm}^2$$

From the results obtained from total surface area of the bodies, we observed that the total surface area of the cone is greater than that of the sphere since $9\pi(\sqrt{17} + 1) > 36\pi$

36. **D.** Note $x^2 + y^2 = (x + y)^2 - 2xy$; $(x + y)$ is not a factor

$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$; $(x + y)$ is not a factor but $(x - y)$ is a factor

$x^5 - y^5 = (x - y)(x^4 + x^3y + x^2y^2 + xy^3 + y^4)$; $(x + y)$ is not a factor but $(x - y)$ is a factor

To now decide if $2x^2 - 3xy + y^2 - x + 1$

and $2x^3 + 2x^2y - xy + 3x - y^2 + 3y$ have $(x + y)$ as factor, we use the factor's theorem as follows:

we set $x + y = 0$ $x = -y$

We then substitute $x = -y$ into the two expressions (or functions) and see which one gives us zero, the one that results into zero is the one that has $x + y$ as it's factor

$$\text{let } f(x, y) = 2x^2 - 3xy + y^2 - x + 1$$

$$f(x = -y, y) = 2(-y)^2 - 3(-y)(y) + y^2 - (-y) + 1$$

$$= 2y^2 + 3y^2 + y^2 + y + 1$$

$$= 6y^2 + y + 1 \neq 0$$

$\therefore (x + y)$ is not a factor of $2x^2 - 3xy + y^2 - x + 1$ next

$$\text{let } f(x, y) = 2x^3 + 2x^2y - xy + 3x - y^2 + 3y$$

$$f(x = -y, y) = 2(-y)^3 + 2(-y)^2y - (-y)y + 3(-y) - y^2 + 3y$$

$$= -2y^3 + 2y^3 + y^2 - 3y - y^2 + 3y$$

$$= 0 + y^2 - y^2 - 3y + 3y$$

$$= 0 + 0 = 0$$

hence $(x - y)$ is a factor of $2x^3 + 2x^2y - xy + 3x - y^2 + 3y$

Remark: (shortcut) For the last two functions tested, to know which is a factor of $x + y$, we pick the one that has no constant (or number) standing alone in it, in the first function we had +1 in the function which ruled it out but the second has no constant in it, which makes it suitable to have $(x + y)$ as it's factor.

37. **B.** Two triangles have the same area when they have the same dimension, in term if sides i.e when the three sides are equal in the triangles.

38. **B.** $(25)^{x-1} = 64 \left(\frac{5}{2}\right)^6$ since $25 = 5^2$; $64 = 2^6$

and from indices, $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

then we have $(5^2)^{x-1} = 2^6 \cdot \frac{5^6}{2^6} \Rightarrow \frac{2^6}{2^6} \cdot 5^6 = 1 \cdot 5^6$

also from indices $(a^n)^m = a^{nm}$

$5^{2(x-1)} = 5^6$ now equating power of 5 gives

$$2(x - 1) = 6;$$

$$2x - 2 = 6$$

$$2x = 6 + 2;$$

$$2x = 8; x = 4$$

39. **A.** Given that $\log_e 4.4 = 1.4816$; $\log_e 7.7 = 2.0142$

$$\log_e 4.4 = \log_e \frac{44}{10} = 1.4816 \log_e 7.7 = \log_e \frac{77}{10} = 2.0142$$

note that from the law of logarithms

$$\log A + \log B = \log AB; \quad \log A - \log B = \log \frac{A}{B}$$

$$\text{now, } \log_e \frac{7}{4} = \log_e \frac{7}{4} \times 1 = \log_e \frac{7}{4} \times \frac{11}{11} = \log_e \frac{77}{44} \times 1$$

$$= \log_e \frac{77}{44} \times \frac{10}{10} = \log_e \frac{77}{10} \times \frac{10}{44}$$

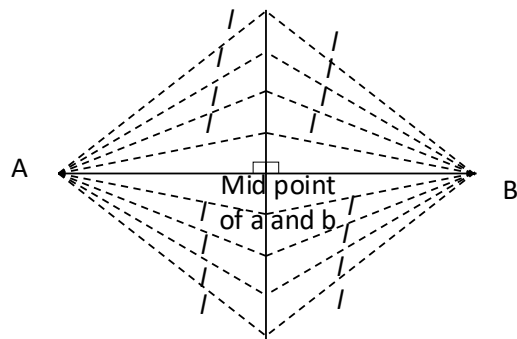
$$= \log_e \frac{77}{10} \div \frac{44}{10} = \log_e 7.7 \div 4.4$$

$$= \log_e \frac{7.7}{4.4} \quad \text{hence } \log_e \frac{7}{4} = \log_e \frac{7.7}{4.4}$$

and from the law of logarithm, we have that

$$\log \frac{A}{B} = \log A - \log B \Rightarrow \log_e \frac{7}{4} = \log_e \frac{7.7}{4.4}$$

$$= \log_e 7.7 - \log_e 4.4 = 2.0142 - 1.4816 = 0.5326$$



40. **B.**

The locus of all points having a distance of 1 units from points A and B is the line perpendicular to the line $|AB|$, passing through the mid point of line $|AB|$.

41. **D.** $\frac{\ln \sqrt{216} - \ln \sqrt{125} - \ln \sqrt{8}}{2(\ln 5 - \ln 3)}$

Note: $\ln()$ is the same as the natural logarithm $\log_e()$ and we shall test the $\ln()$ as $\log_e()$ as follows and using the following laws of logarithms and indices

$$\ln A - \ln B = \ln \frac{A}{B}; \sqrt{a^n} = a^{\frac{n}{2}}; \ln A^x = x \ln A; a^{\frac{n}{m}} = (\sqrt[m]{a})^n$$

now,

$$\sqrt{216} = \sqrt{6^3} = 6^{\frac{3}{2}}; \quad \sqrt{8} = \sqrt{2^3} = 2^{\frac{3}{2}}; \quad \sqrt{125} = \sqrt{5^3} = 5^{\frac{3}{2}}$$

$$\text{so that } \frac{\ln \sqrt{216} - \ln \sqrt{125} - \ln \sqrt{8}}{2(\ln 5 - \ln 3)} = \frac{\ln 6^{\frac{3}{2}} - \ln 5^{\frac{3}{2}} - \ln 2^{\frac{3}{2}}}{2 \ln \frac{5}{3}}$$

$$= \frac{\frac{3}{2} \ln 6 - \frac{3}{2} \ln 5 - \frac{3}{2} \ln 2}{2 \ln \frac{5}{3}} = \frac{\frac{3}{2} [\ln 6 - \ln 5 - \ln 2]}{2 \ln \frac{5}{3}}$$

$$= \frac{\frac{3}{2} [\ln \frac{6}{5 \times 2}]}{2 \ln \frac{5}{3}} = \frac{\frac{3}{2} [\ln (\frac{3}{5})^1]}{2 \ln (\frac{5}{3})^1}$$

$$\text{also from laws of indices } \left(\frac{a}{b}\right)^1 = \left(\frac{b}{a}\right)^{-1}$$

$$\text{so that we have } \frac{\ln \sqrt{216} - \ln \sqrt{125} - \ln \sqrt{8}}{2(\ln 5 - \ln 3)} = \frac{\frac{3}{2} [\ln (\frac{3}{5})^{-1}]}{2 \ln (\frac{5}{3})^1} = \frac{-\frac{3}{2} [\ln (\frac{5}{3})]}{2 \ln (\frac{5}{3})}$$

$$= -\frac{3}{2} \div 2 = -\frac{3}{2} \times \frac{1}{2} = -\frac{3}{4}$$

$$\text{Hence: } \frac{\ln \sqrt{216} - \ln \sqrt{125} - \ln \sqrt{8}}{2(\ln 5 - \ln 3)} = -\frac{3}{4}$$

42. **E.** $\frac{(a^{-\frac{1}{a}})(a^{\frac{4}{3} + \frac{1}{2}})}{(a^2 - \frac{1}{a^2})}$

From indices (i) $a^n \times a^m = a^{n+m}$;

$$(ii) a^n \div a^m = a^{n-m}; \quad (iii) \frac{1}{a^m} = a^{-m}$$

$$\frac{(a^{-\frac{1}{a}})(a^{\frac{4}{3} + \frac{1}{2}})}{(a^2 - \frac{1}{a^2})} = \frac{(a^{-a^{-1}})(a^{\frac{4}{3} + a^{-\frac{2}{3}}})}{(a^2 - a^{-2})}$$

$$\begin{aligned}
&= \frac{a^{1 \times \frac{4}{3} + a^{1 \times \frac{-2}{3}} - a^{-1 \times \frac{4}{3}} - a^{-1 \times \frac{-2}{3}}}{(a^2 - a^{-2})} \\
&= \frac{a^{\frac{7}{3} + a^{\frac{1}{3}} - a^{\frac{1}{3}} - a^{\frac{-5}{3}}}{a^{-2}(a^4 - 1)} \\
&= \frac{a^{\frac{7}{3} - a^{\frac{-5}{3}}}}{a^{-2}(a^4 - 1)} = \frac{a^{-\frac{5}{3}}(a^4 - 1)}{a^{-2}(a^4 - 1)} = a^{-\frac{5}{3}} \div a^{-2} \\
&= a^{-\frac{5}{3} - (-2)} = a^{-\frac{5}{3} + 2} = a^{\frac{1}{3}} \\
&\text{hence, } \frac{(a^{-\frac{1}{a}})(a^{\frac{4}{3} + \frac{1}{a^{\frac{2}{3}}}})}{(a^2 - \frac{1}{a^2})} = a^{\frac{1}{3}}
\end{aligned}$$

43. A. note factors theorem: if $P(x)$ is a polynomial that is divisible by $x - a$ then $P(x = a) = 0$.

Remainder's theorem: if $P(x)$ is a polynomial that when divided by $x - b$, it leaves a remainder, then

$P(x = b) = \text{remainder}$

Now $p(x) = x^4 - kx^3 + 10x^2 + lx - 3$ is divisible by $(x - 1)$,

Using factor's theorem, we have that

$$P(x = 1) = 1^4 - k(1)^3 + 10(1)^2 + l(1) - 3 = 0;$$

$$\Rightarrow 1 - k + 10 + l - 3 = 0$$

$$k - l = 8 \quad \text{--- (1) also,}$$

$$p(x) = x^4 - kx^3 + 10x^2 + lx - 3$$

when divided by $x + 2$ leaves a remainder 27, so using the remainder's theorem we have that

$$p(x = -2) = (-2)^4 - k(-2)^3 + 10(-2)^2 + l(-2) - 3 = 27$$

$$16 + 8k + 40 - 2l - 3 = 27$$

$$8k - 2l = 27 + 3 - 16 - 40$$

$$8k - 2l = -26$$

$$\Rightarrow 4k - l = -13 \quad \text{--- (2)}$$

$$\text{from (1) } k - l = 8 \quad \text{--- (1)}$$

This is a simultaneous equation and we solve by using the elimination method as follows: subtract (1) from (2) to have

$$3k - 0 = -13 - 8 \quad 3k = -21$$

$$k = -\frac{21}{3} = -7; \quad \text{substitute } k = -7 \text{ into (1)}$$

$$k - l = 8; \quad -7 - l = 8$$

$$-7 - 8 = l, \quad l = -15$$

$$\text{hence, } k = -7, \quad l = -15$$

44. E. considering the quadrant table below

2 nd Quadrant (90° - 180°)	(0° - 90°) 1 st Quadrant
$\sin(180^\circ - \theta) = \sin \theta$	$\sin \theta$
$\cos(180^\circ - \theta) = -\cos \theta$	$\cos \theta$
$\tan(180^\circ - \theta) = -\tan \theta$	$\tan \theta$
$\sin(270^\circ - \theta) = -\sin \theta$	$\sin(360^\circ - \theta) = -\sin \theta$
$\cos(270^\circ - \theta) = -\cos \theta$	$\cos(360^\circ - \theta) = \cos \theta$
$\tan(270^\circ - \theta) = \tan \theta$	$\tan(360^\circ - \theta) = -\tan \theta$
3 rd Quadrant (180° - 270°)	(270° - 360°) 4 th Quadrant

From the table we observe that sine of angles is negative in the 3rd and 4th quadrant where the angles will be $270^\circ - \theta$ and $360^\circ - \theta$ when θ is an angle between 0° and 360° . Note that the whole quadrants covers only angles from 0° to 360° . When solving angles more than this range we subtract multiples of 360° from such angle but when solving angles less than the range we add multiples of 360° to it, so as to adjust the angle to the range $(0^\circ - 360^\circ)$

Now for $\sin -1290^\circ$;

-1290° is less than the range $(0^\circ - 360^\circ)$ so we add multiples of 360° to it so that we have

$$\sin(-1290^\circ) = \sin(-1290^\circ + 360^\circ) = \sin(-930^\circ)$$

$$= (-930^\circ + 360^\circ) = \sin(-570^\circ)$$

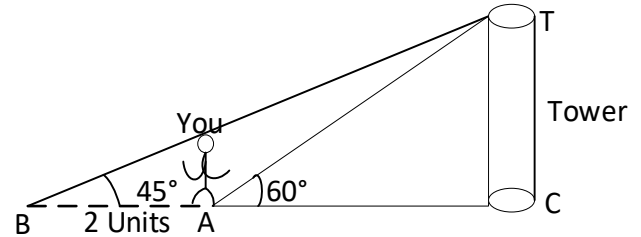
$$= \sin(-570^\circ + 360^\circ) = \sin(-210^\circ)$$

$$\sin(-1290^\circ) = \sin(-210^\circ + 360^\circ) = \sin(150^\circ)$$

Observe that 150° is in the second quadrant where sine of angles is positive so we have $\sin(-1290^\circ) = \sin 150^\circ$

$$= +\sin(180^\circ - 150^\circ) = +\sin 30^\circ = +\frac{1}{2} = \frac{1}{2}$$

45. D.



We are to find the distance $|AC|$

observe that $|BC| = 2 \text{ units} + |AC|$

Now considering triangle ΔTBC ;

$$\frac{\tan 45^\circ}{1} = \frac{|TC|}{|BC|} = \frac{|TC|}{2 + |AC|} = \tan 45^\circ$$

$$|TC| = \tan 45^\circ (2 + |AC|) = 1(2 + |AC|)$$

$$|TC| = 2 + |AC| \quad \text{--- (1)}$$

also considering triangle ΔTAC

$$\frac{\tan 60^\circ}{1} = \frac{|TC|}{|AC|} \quad \text{on cross multiplying}$$

$$|AC| \tan 60^\circ = |TC| \quad |TC| = |AC| \cdot \tan 60^\circ$$

$$|TC| = |AC| \cdot \sqrt{3} \quad \text{--- (2) from equation (1) and (2)}$$

$$|TC| = 2 + |AC| = |AC| \cdot \sqrt{3}$$

$$2 + |AC| = |AC| \sqrt{3}; \quad |AC| \sqrt{3} - |AC| \cdot 1 = 2$$

$$|AC|(\sqrt{3} - 1) = 2; \quad |AC| = \frac{2}{\sqrt{3} - 1}$$

$$\text{rationalizing the surd gives } |AC| = \frac{2}{\sqrt{3} - 1} \cdot \frac{\sqrt{3} + 1}{\sqrt{3} + 1} =$$

$$\frac{2\sqrt{3} + 2}{\sqrt{9} + \sqrt{3} - \sqrt{3} - 1} = \frac{2(\sqrt{3} + 1)}{3 - 1} = \frac{2(\sqrt{3} + 1)}{2}$$

$$\text{Hence, } |AC| = (\sqrt{3} + 1) \text{ units; } |AC| = 1 + \sqrt{3} \text{ units}$$